



Lab02

Frequency Divider

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Outline

- Sequential Circuit
 - Waveform Difference
- Frequency Divider (Lab02)
 - FreqDiv.sv
 - SPEC
 - Hint
- Grading Policy
 - Upload

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Sequential Circuit

- Different from combinational circuit, sequential circuit will synthesize registers which can store previous data.

```
always @(posedge clk or negedge rst_n) begin
    if(!rst_n)
        a <= 0;
    else
        a <= a + 1;
end
```

← Sequential circuit

```
assign a = a + 1;

always @(*) begin
    a = a + 1;
end

always @(*) begin
    if(a > b) a = b;
    else a = a;
end
```

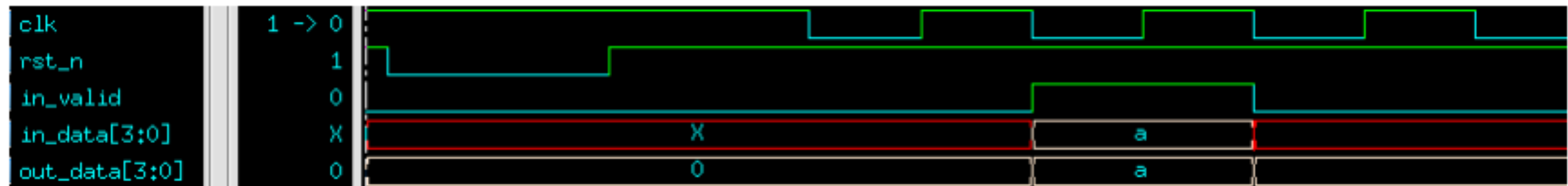
← Combinational circuit

Waveform Difference

- Combinational Circuit

```
always@(*)begin
    if(in_valid) out_data = in_data;
    else out_data = 0;
end

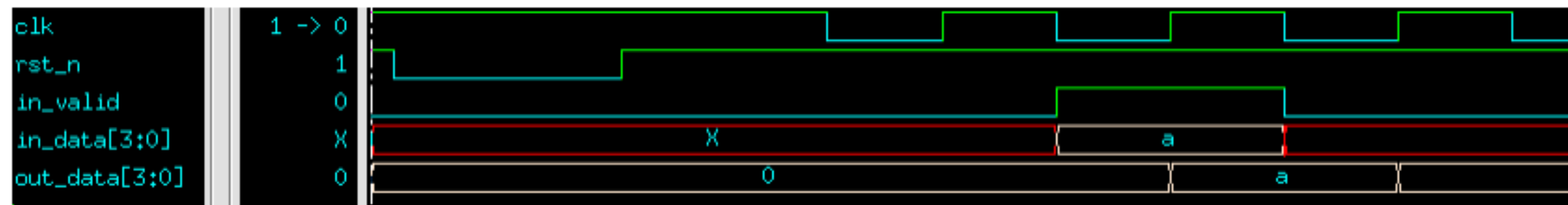
assign out_data = (in_valid) ? in_data : 0;
```



Waveform Difference

- Sequential Circuit

```
always@(posedge clk or negedge rst_n)begin
    if(!rst_n)
        out_data <= 0;
    else begin
        if(in_valid) out_data <= in_data;
        else out_data <= 0;
    end
end
```

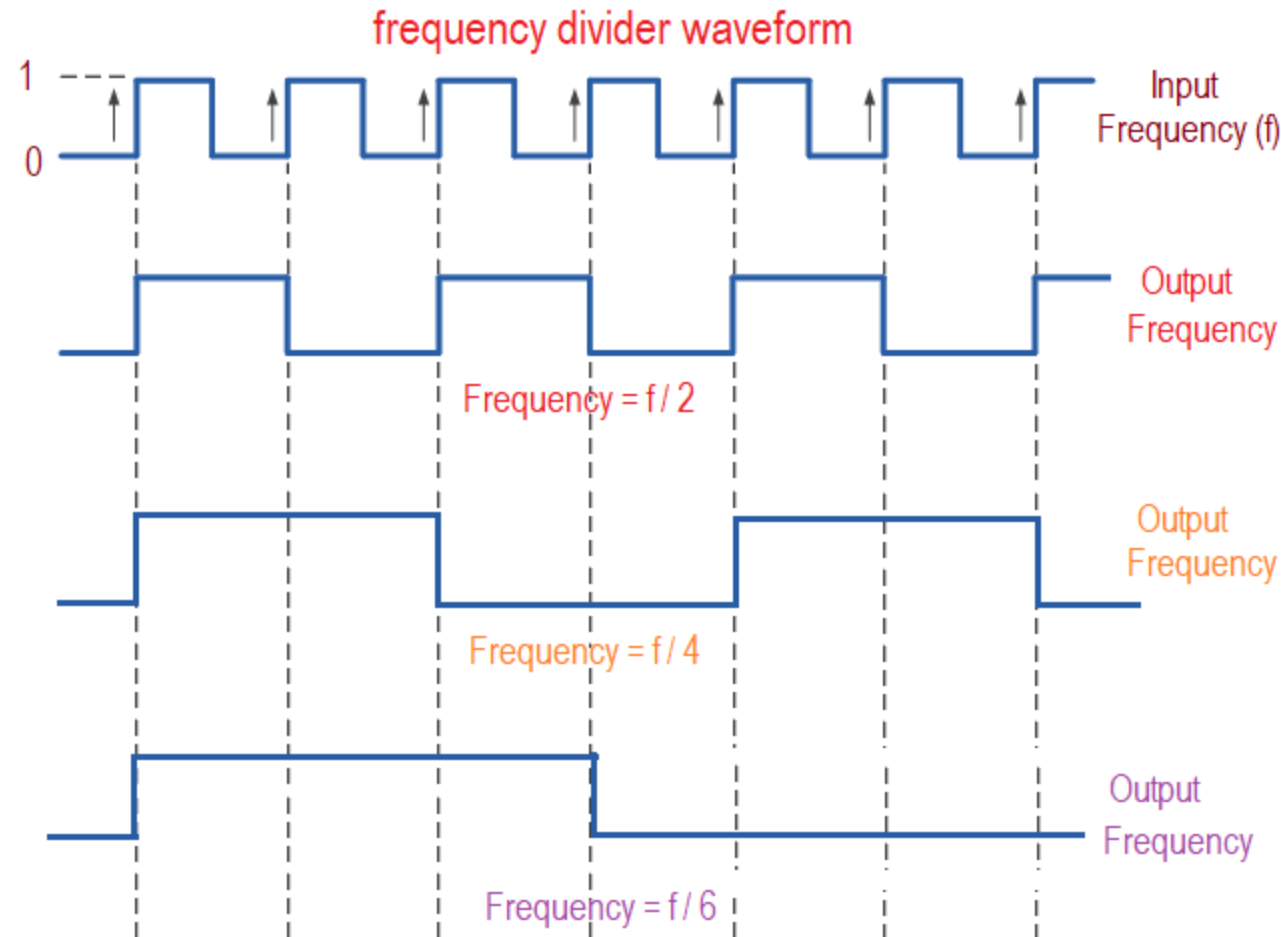


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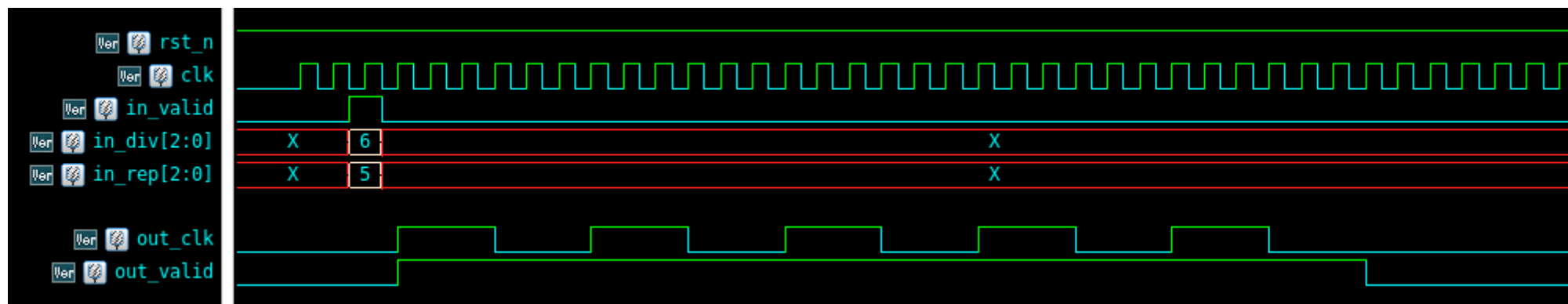
Frequency Divider (Lab02)

- FreqDiv.sv
 - 除頻器可以將高頻 clk 訊號轉換為低頻的 out_clk



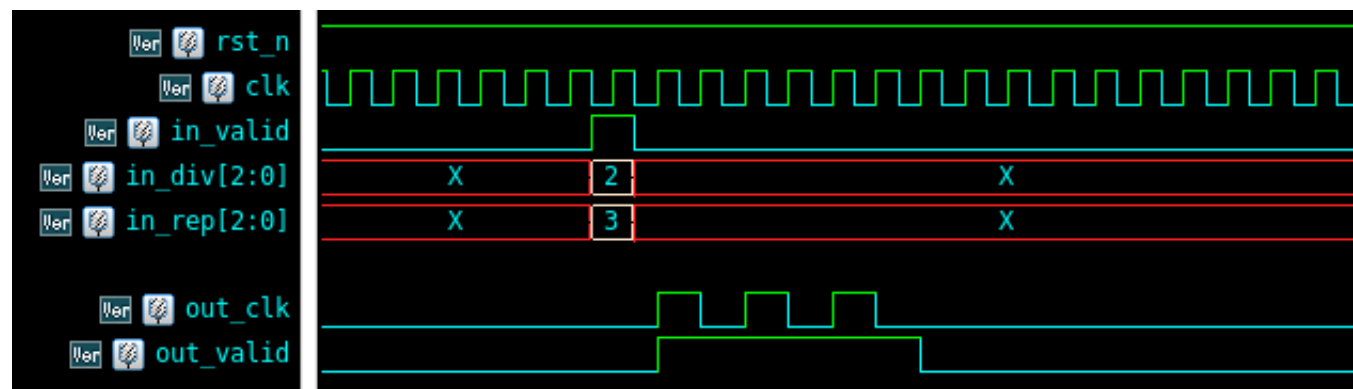
Frequency Divider (Lab02)

- FreqDiv.sv
 - 本次 Lab 將實作偶數除頻器 (`in_div` 僅會出現 2、4、6 的 case)
 - 輸出的波型必須重複指定次數 (`in_rep` 範圍為 1 ~ 7)
 - 標註有效波型範圍 (`out_valid`)
- Ex.1
 - 將 `clk` 除以 6，並重複輸出 5 次



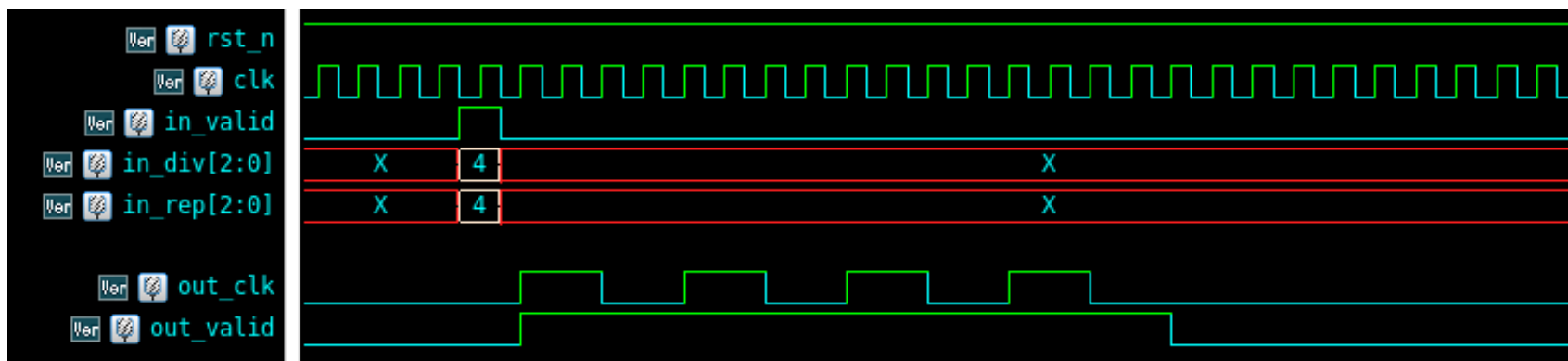
Frequency Divider (Lab02)

- FreqDiv.sv
 - 本次 Lab 將實作偶數除頻器 (`in_div` 僅會出現 2、4、6 的 case)
 - 輸出的波型必須重複指定次數 (`in_rep` 範圍為 1 ~ 7)
 - 標註有效波型範圍 (`out_valid`)
- Ex.2
 - 將 `clk` 除以 2，並重複輸出 3 次



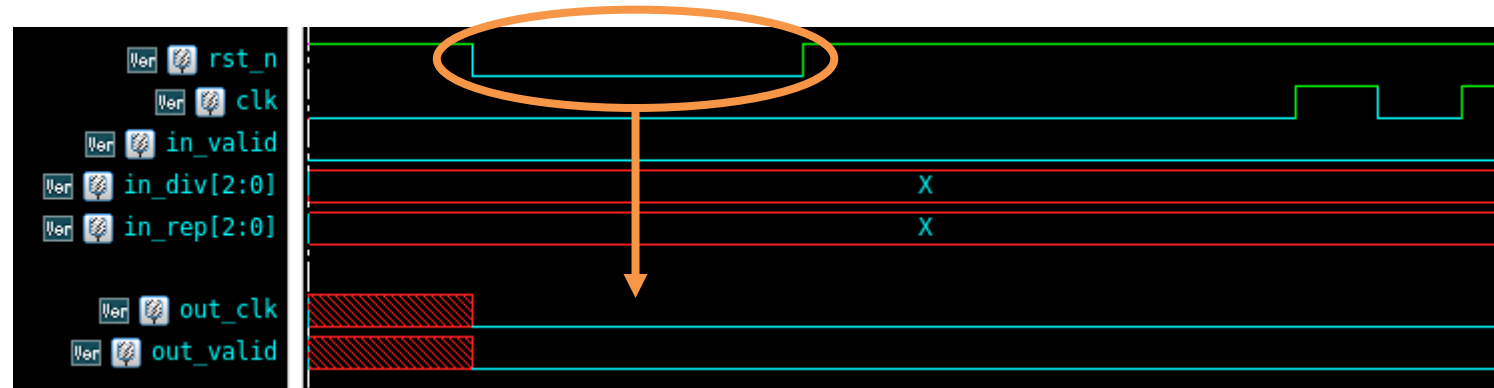
Frequency Divider (Lab02)

- FreqDiv.sv
 - 本次 Lab 將實作偶數除頻器 (`in_div` 僅會出現 2、4、6 的 case)
 - 輸出的波型必須重複指定次數 (`in_rep` 範圍為 1 ~ 7)
 - 標註有效波型範圍 (`out_valid`)
- Ex.3
 - 將 `clk` 除以 4，並重複輸出 4 次

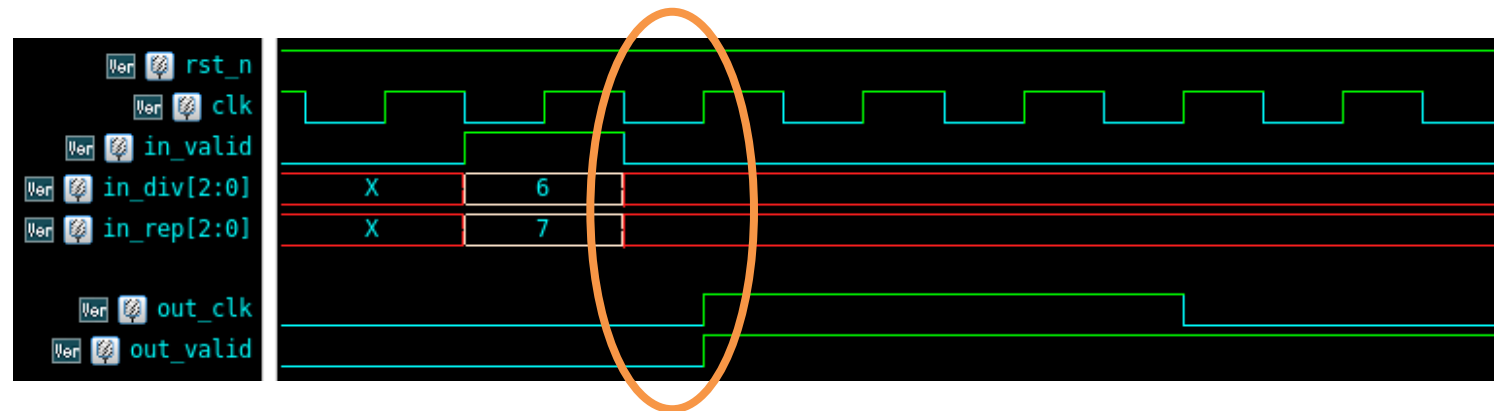


Frequency Divider (Lab02)

- FreqDiv.sv
 - Output should be reset after “rst_n”

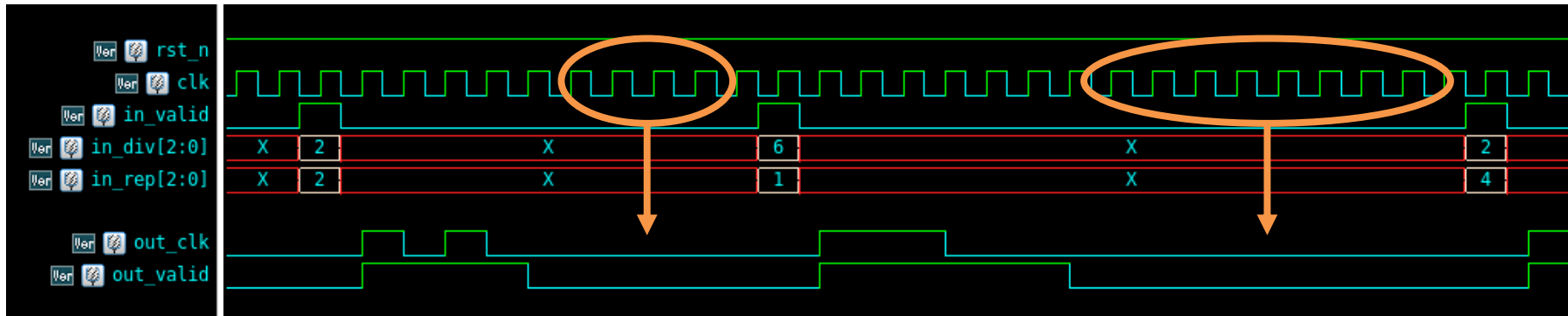


- “out_valid” shouldn't overlap with “in_valid”

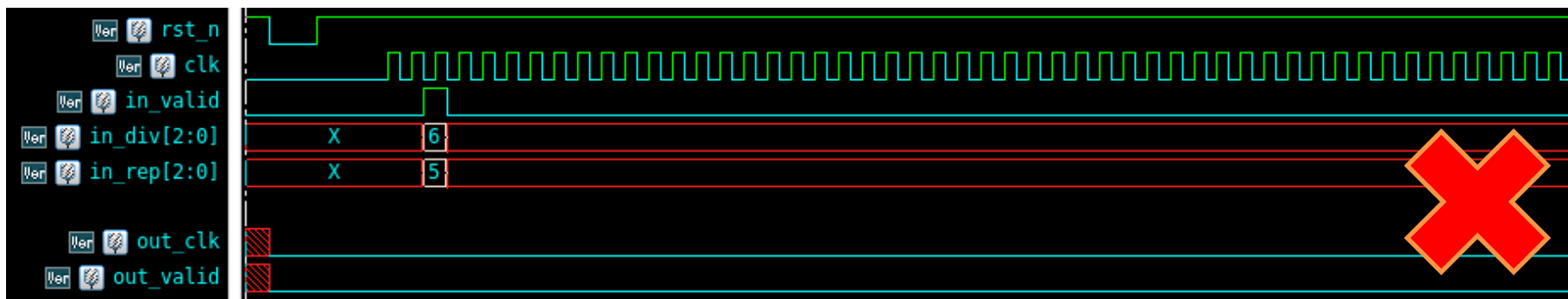


Frequency Divider (Lab02)

- FreqDiv.sv
 - The next input will arrive within 5 to 10 cycles (output should stay 0).



- “out_valid” should be asserted within 100 cycles after “in_valid”.



Frequency Divider (Lab02)

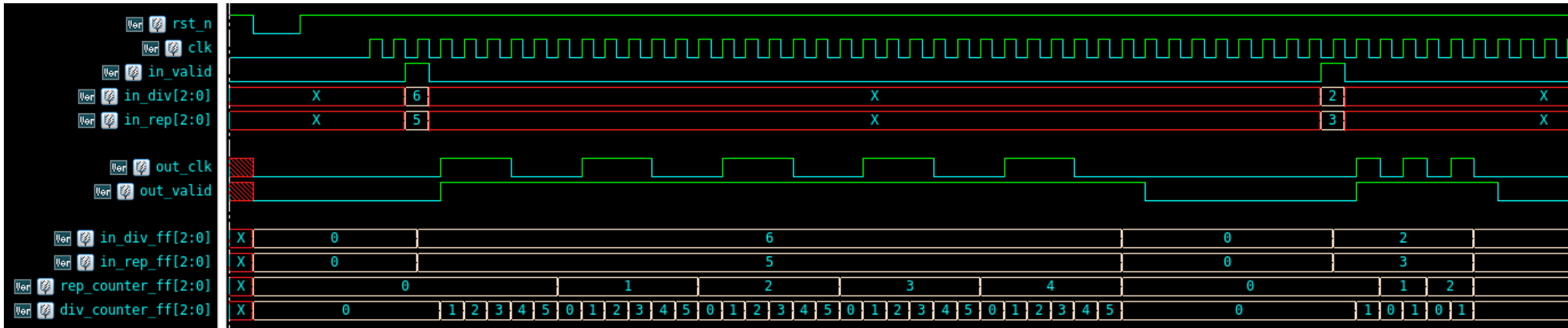
- SPEC

Input Signal	Bit Width	Definition
rst_n	1	Asynchronous active-low reset
clk	1	Global clock
in_valid	1	Set to 1 when the input is ready
in_div	3	Frequency division factor (2, 4, or 6)
in_rep	3	Number of output repetitions (1 ~ 7)

Output Signal	Bit Width	Definition
out_clk	1	Output clock
out_valid	1	Set to 1 when the output is ready

Frequency Divider (Lab02)

- Hint
 - 使用兩層 counter 來設計電路
 - 外層 rep_counter_ff 計算 in_rep (ex. 0、1、2、3、4)
 - 內層 div_counter_ff 計算 in_div (ex. 0、1、2、3、4、5、0、1、2、3、4、5、...)
 - 內層 div_counter_ff 小於 in_div_ff 的一半時 “out_clk” 為 1；反之 “out_clk” 為 0
 - 當內外層 counter 都還沒數到盡頭時 “out_valid” 持續為 1



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Grading Policy

- 1st demo – 100% for following requirements:
 - Pass the RTL simulation
 - Pass Synthesis (No **Error**, Timing Report Slack **MET**, No **Latch**)
 - Pass Gate-level simulation
- 2nd demo – 60%

Upload

- `cd ../09_UPLOAD/`
- `./01_upload`
- `./02_download [argument]`
 - `[argument] = demo1 or demo2`
- Note:
 - Make sure to upload the correct file.
 - Deadline:
 - 1st demo: 3/12 (四) 17:30
 - 2nd demo: 3/12 (四) 23:59